

BORJA JAVIERRE MOYANO

Senior Video Systems & Real-Time Communications Engineer

Barcelona, Spain (relocating from Stockholm, September 2026)

EU citizen – no visa sponsorship required

+46 76 451 40 94 | borja.javierre@gmail.com

linkedin.com/in/borjajavierre | github.com/jackiefasty | jackiefasty.github.io



PROFESSIONAL SUMMARY

Senior software engineer with **3 years embedded in Google Meet's Video team**, tuning C++/Python media pipelines, **VP8/VP9/AV1** codecs, and **WebRTC** internals for a product serving **hundreds of millions of users**. 5+ years total across real-time media, streaming, and backend infrastructure, spanning **FFmpeg/GStreamer** pipelines, automated codec regression testing, and ML-based video quality prediction. **Promoted to Senior Engineer after 16 months at Google**. Fluent in modern AI-assisted development (LLMs, MCP servers, RAG, agentic workflows) as a delivery multiplier.

CORE SKILLS

Video & Media: WebRTC, libwebRTC, RTP/RTCP, SDP, jitter buffer, bandwidth estimation, congestion control, simulcast, VP8, VP9, AV1, H.264, transcoding, containers, streaming protocols, FFmpeg, GStreamer, latency optimisation, QoE, video quality metrics

Languages: C++, Python, TypeScript, JavaScript, SQL, C, C#

Testing & CI/CD: Automated regression testing, codec test harnesses, Jenkins, GitHub Actions, CI/CD pipelines, API testing, Docker, Linux

Backend & Cloud: FastAPI, REST APIs, microservices, PostgreSQL, GCP (Cloud Run, Firestore, Pub/Sub, IAM), Terraform, Infrastructure as Code, AWS (familiar)

ML / Data: Model parameter tuning, video quality prediction, Python data stack, SQL, predictive modelling

AI-Assisted Development: LLMs, MCP servers, RAG pipelines, agentic workflows, Claude Code, Codex, prompt engineering

Ways of Working: Agile, Scrum, sprint planning, OKR frameworks, technical design documents, code review, mentoring, stakeholder management, Jira, Confluence

PROFESSIONAL EXPERIENCE

Senior Software Engineer (promoted from Software Engineer) – HCLTech Sweden AB / Codemill AB – embedded consultant for Google Sweden AB, Stockholm | Sep 2023 – May 2026

Embedded engineer in the Google Meet Video team, working directly on real-time video quality at global scale. Promoted to Senior after 16 months.

- Improved compression efficiency and end-to-end latency in Google Meet for **hundreds of millions of users**, by running codec testing and parameter tuning across **VP8, VP9, and AV1** against objective video quality targets.
- Cut video codec test cycle time by **30%** and expanded coverage with **50+** new test cases spanning VP8 and AV1, by designing and automating the regression testing pipeline in Python.
- Improved real-time communication reliability under packet loss and constrained bandwidth, by tuning **WebRTC** internals – bandwidth estimation, jitter buffer behaviour, and congestion control – in production C++ code.
- Enabled reliable media processing and format handling across the team's test and analysis tooling, by integrating **FFmpeg and GStreamer** pipelines for transcoding, container handling, and stream ingest.
- Reached **87–92% accuracy** in video quality prediction, by tuning ML model parameters and rebuilding the training and evaluation loop in Python.
- Drove OKR planning for H1/H2 2025 across **4 concurrent projects**, by running 1:1s and weekly syncs and authoring quarterly and annual reviews presented to Google stakeholders.
- Reduced ramp-up time for new team members, by driving recruitment and onboarding and mentoring **2 junior engineers** through their first 3 months.

Senior Software Engineer – Elvy AB (Geofy), Stockholm, Sweden | May 2026 – Aug 2026

Backend, infrastructure, and full-stack ownership at an early-stage energy subscription provider. Role eliminated in a company-wide budget reduction.

- Delivered a production installer portal plus adjacent admin dashboards in **4 weeks**, onboarding **30+ installers**, by owning technical design through release across **4 repositories** (**Next.js** frontends, **Python/FastAPI** backend) with role-scoped **JWT** authorisation and a lead → approval → credential-issuance flow validated in a 2-installer pilot.
- Eliminated broad service-account access from client applications following a security review, by moving password hashing and datastore writes behind a single authenticated server-side REST endpoint.

- Migrated communications and installer data from **Firestore** to a **10+ table PostgreSQL** schema on **Cloud SQL** with **no downtime**, by designing the relational model around a shared person entity and building one-time backfill tooling, and managed all **GCP** infrastructure (Cloud Run, IAM, Firestore) as code with **Terraform**.
- Reduced feature lead time on multi-repository work by an estimated **20+ engineering hours per month**, by designing a **multi-agent AI development workflow** (LLMs, **MCP servers**, **RAG** retrieval, specialised planner/coder/reviewer agents) and codifying it as a shared team engineering standard.

Test Engineer Intern – Agile Content / Edgeware AB, Stockholm, Sweden | Jun 2022 – Oct 2022

- Reduced manual test execution by **20%** for C++ media streaming components, by writing automated Python test cases integrated into Jenkins CI.

Teaching Assistant (part-time) – KTH Royal Institute of Technology, Stockholm, Sweden | Mar 2023 – Sep 2024

- Supported **approximately 30 master's students** per semester in Computer Graphics (C++, C#) and Database Technology (Python), by running lab sessions and delivering structured code review feedback.

Data Scientist Intern (Master's Thesis) – Telenor Sweden AB, Stockholm, Sweden | Jan 2023 – Sep 2023

- Cut average diagnosis effort for TV support teams by **74–86%** depending on model, by building and evaluating ML models in Python/SQL on AWS that predict case resolution time from CRM ticket history.

Automation Testing & QA Engineer – Worldline Global S.A., Barcelona, Spain | Feb 2021 – Sep 2021

- Cut manual regression testing time by **30%** across payment APIs and mobile clients, by automating test suites in Cucumber (Ruby) and Python.
- Caught **20+ critical defects** before release, by building Postman-based API validation and regression coverage for payment flows.

Research Intern – CTTC / Universitat Politècnica de Catalunya, Castelldefels, Spain | Dec 2019 – Jul 2020

- Evaluated interoperability across open-source Intelligent Transport Systems platforms as Bachelor's thesis work, by designing the comparison methodology and running the test campaign.

Broadcast Engineering Intern – Cellnex Telecom S.A., Barcelona, Spain | Mar 2019 – Sep 2019

- Delivered RF and signal coverage calculations across **350+ broadcast stations** for Spain's Second Digital Dividend frequency migration, supporting DTT emission legalisation projects.

SELECTED PROJECTS & PUBLICATIONS

- **WebRTC Playground** – Static two-peer **WebRTC** test suite simulating sender/receiver calls in-page via paired **RTCPeerConnections** with local ICE/SDP exchange. Adds and removes **simulcast** layers, tunes per-layer maxBitrate, scaleResolutionDownBy and scalabilityMode, forces codecs across **VP8/VP9/AV1/H.264**, and polls **getStats()** for per-layer bitrate, framerate, QP and encoder time. Live demos on GitHub Pages. github.com/jackiefasty/webrtc-playground
- **Multi-Agent AI Development Framework** – Built a role-specialised agent system (planner, backend/frontend coder, tester, reviewer) on top of an LLM CLI, with **MCP server** integrations, **RAG** retrieval, and a codified behavioural specification; adopted as a shared team engineering standard. Python, TypeScript.
- **ITS Platform Interoperability Evaluation (CTTC / Universitat Politècnica de Catalunya)** – Comparative evaluation of open-source Intelligent Transport Systems stacks; findings contributed to a research publication.

EDUCATION

MEng, ICT Innovation – Visual Computing Track | KTH Royal Institute of Technology, Stockholm, Sweden | 2021–2023

BEng, Audiovisual Systems Engineering of Telecommunications | Universitat Pompeu Fabra, Barcelona, Spain | 2015–2020

LANGUAGES & ADDITIONAL

Languages: Spanish (native) | Catalan (native) | English (C1, professional working proficiency) | Swedish (B1) | French (A1)

Additional: EU driving licence (Type B) | International volunteering: Berlin, Stuttgart, Warsaw